

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessor and ALP

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain fetch, decode and execute. What are the major differences between microprocessor and microcontroller? 7
b) Define addressing mode. Explain different addressing modes of 8085 with example. 8
2. a) Draw and explain the architecture of the 8085 microprocessor with a neat block diagram. 7
b) Write an assembly language program in 8085 to transfer 10 bytes of data from 3020H to 3030H if the data is greater than 4FH. 8
3. a) Why are flags in microprocessor necessary? Explain it in the context of 8086. 7
b) Draw and explain the functional diagram of 8086 microprocessor. 8
4. a) Write a MASM program for 8086 to display "POKHARA UNIVERSITY" in reverse order. 7

OR

Write a MASM program for 8086 which transfer 10 bytes of data stored in "array 1" to the new location "array 2".

- b) Design an address decoding circuit to interface two 2KB RAM and one 4KB ROM with initial address of your choice. 8
5. a) Draw the architecture of 8255 – PPI controller and explain its modes of operation. 7

OR

Draw and explain the architecture of 8251 USART controller.

- b) Why do we use Intel 8237A DMA controller? Explain with its architecture. 8
6. a) Explain the difference between polled and vectored interrupt handling technique with necessary diagram. 7

b) How multiple interrupt can be handled using priority interrupt controller (PIC). Explain 8259 PIC with necessary block diagram. 8

7. Write short notes on: (Any two) 2×5

- a) Timing diagram of MOV M,A
- b) I/O mapped I/O vs memory mapped I/O
- c) Interrupt vector table